EE3400 – Tutorial 3

Integrated & Problem-Solving Tutorial: Sensors, Strain Gauges and Instrumentation Amplifiers

Purpose: This tutorial is intentionally more demanding than Tutorial 2. The questions require students to connect concepts, make engineering decisions, identify hidden problems and justify their answers. Use the equations and terminology developed in Lectures 1–3.

Instructions

Show all calculations. Where more than one engineering solution is possible, state your assumptions and justify your choice. Do not simply quote a formula: explain what the result means physically.

QUESTION 1 – From Physical Quantity to Electrical Signal

A pressure vessel is fitted with a bonded strain gauge. The pressure changes the vessel wall stress, which changes the strain gauge resistance. The resistance change is then converted into a small differential voltage by a Wheatstone bridge.

(a) Identify the *primary, secondary and tertiary measurands* in this measurement chain.

(b) Explain why the strain gauge does not directly measure pressure.

(c) Give two reasons why a Wheatstone bridge is appropriate for this application.

(d) The bridge produces only a few millivolts. Explain why connecting a low-resistance measuring device directly to the bridge could cause an error.

QUESTION 2 – Sensor Specification & Calibration

A temperature sensor is specified to produce 0.20 V at 0 °C and 4.80 V at 100 °C. Assume a linear transfer characteristic.

(a) Determine the sensor sensitivity in V/°C and mV/°C.

(b) During a test, the sensor output is 3.42 V. Estimate the temperature indicated by the sensor.

(c) A calibrated thermometer shows that the actual temperature is 72 °C. Calculate the measurement error and percentage error of the sensor reading from (b).

(d) The sensor will be used to measure a rapidly changing temperature. Explain why bandwidth must be considered even if the sensor has excellent accuracy.

QUESTION 3 – The Nyquist Trap

A sensor produces a useful signal containing frequencies from DC to 20 Hz. Unfortunately, a strong 50 Hz mains interference is also present. The signal is to be digitised by an ADC.

(a) What is the minimum theoretical sampling frequency required by the Nyquist criterion for the useful signal?

(b) Would simply choosing a 40 Hz sampling frequency solve the problem? Explain carefully.

(c) You decide to sample at 200 Hz. Propose a suitable filtering strategy for the 50 Hz interference and explain why a simple low-pass filter with a 20 Hz cutoff may be preferable to doing no filtering.

(d) Explain what could happen if the 50 Hz component is not adequately removed before sampling.

QUESTION 4 – Strain Gauge: What Really Changes?

A metallic strain gauge has an initial resistance of 350 Ω and a gauge factor of 2.0. It is bonded to a steel member subjected to a tensile strain of 800 με.

(a) Calculate the change in resistance caused by the strain.

(b) Calculate the new resistance.

(c) The temperature coefficient of resistance of the gauge is 0.0005/°C. The temperature rises by 20 °C. Calculate the resistance change due to temperature alone.

(d) Compare the resistance changes from strain and temperature. What does this tell you about temperature compensation in strain-gauge measurements?

QUESTION 5 – Wheatstone Bridge Diagnosis

A quarter-bridge strain-gauge circuit uses four nominally equal resistors of 120 Ω and an excitation voltage of 5 V. One active gauge increases from 120.00 Ω to 120.24 Ω when the member is strained. Assume the other three resistors remain at 120 Ω.

(a) Calculate the two bridge midpoint voltages before and after the strain is applied.

(b) Determine the magnitude and polarity of the differential bridge output. Clearly state your assumed bridge orientation.

(c) Explain why the bridge output is much smaller than the 5 V excitation.

(d) A student says: “The gauge resistance changed by only 0.2%, so the strain signal is useless.” Explain why this conclusion is incorrect.

QUESTION 6 – Instrumentation Amplifier and the Hidden Common-Mode Signal

A strain-gauge bridge produces a differential signal of 1.20 mV while both input terminals sit at approximately 0.80 V relative to ground. A 3-op-amp instrumentation amplifier is used. Let R1 = R3 = 10 kΩ, R2 = R4 = 10 kΩ and R5 = R6 = 25 kΩ, with the classic gain equation:
$\mathbf{V_{OUT} = (V_{IN2} - V_{IN1})\,[1 + 2R5/RG]\,(R2/R1)}$.

(a) Determine the differential gain.

(b) Calculate the output due to the 1.20 mV differential signal.

(c) The instrumentation amplifier has a CMRR of 100 dB. Determine the corresponding linear CMRR.

(d) Using the differential gain from (a), estimate the common-mode gain and the unwanted output produced by the 0.80 V common-mode signal.

(e) Explain why a high CMRR is particularly important in this application.

QUESTION 7 – Design the Complete Measurement Chain

You are asked to design a simple measurement system for a strain-based force sensor. At maximum force, the bridge output is +10 mV relative to its zero-force output. At minimum force, it can be –10 mV.

The ADC accepts 0–3.3 V. You want the zero-force output to be approximately 1.65 V so that the full positive and negative signal can be measured without clipping.

Use a 3-op-amp instrumentation amplifier with R1 = R3 = 10 kΩ, R2 = R4 = 10 kΩ and R5 = R6 = 10 kΩ. The gain equation is:
G = [1 + 2R5/RG](R2/R1).

(a) Choose a practical amplifier gain so that ±10 mV can fit within the ADC input range when a 1.65 V reference is used.

(b) Calculate the required R_G for your chosen gain.

(c) Calculate the expected ADC input voltage at −10 mV, 0 mV and +10 mV.

(d) Explain the role of the reference input in this design.

(e) If the amplifier gain is made unnecessarily large, what failure can occur even though the sensor signal is very small?

QUESTION 8 – Troubleshooting: The System Works… But the Reading Is Wrong

A student builds the following chain:
Strain gauge → Wheatstone bridge → instrumentation amplifier → ADC.
With no mechanical load, the ADC reading slowly changes as the laboratory warms up. In addition, a periodic ripple is visible on the reading. When a hand is placed near the circuit, the ripple becomes larger.

The student proposes replacing the strain gauge with a gauge having a much higher gauge factor. You are not convinced.

(a) Identify at least three possible causes of the observed behaviour using concepts from the lectures.

(b) For each cause, propose a corresponding engineering remedy.

(c) Explain why increasing the gauge factor may improve sensitivity but does not necessarily solve the underlying problem.

(d) Which parts of the signal-conditioning chain—buffering, amplification and filtering—would you consider most important, and why?

QUESTION 9 – Stretch Challenge: Choose the Gauge Arrangement

You must measure deformation on a mechanical component. In one application the strain is predominantly in one known direction. In another, the principal stress direction is unknown. In a third, the strain gradient is very large and you need several measurements at essentially the same location.

(a) For each application, choose between a uniaxial gauge, a bi-axial rosette, a tri-axial rosette, a planar rosette or a stacked rosette.

(b) Justify each choice using the criteria from the lecture rather than simply naming the gauge.

(c) Which choice would you make if heat dissipation is a major concern? Explain.

End of Tutorial 3

The final question is deliberately open-ended. In an engineering problem, a well-justified alternative can be correct when it is supported by sound reasoning.